

WATERWATCH · REQUIREMENTS

WaterWatch requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	R7 and R12 figures from WWT-CAL-001 v0.4 (status light); no status changed
0.5	2026-10-02	Amish Chadha	Status from WWT-CAL-001 v0.3 for the constructable design (WWT-DDR-003); R12 stated against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	For TRL 3, status of every requirement from WWT-CAL-001; decisions from WWT-DDR-001 recorded; targets unchanged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWatch requirements

These requirements were first set for the concept and are checked by calculation in WWT-CAL-001 v0.4. Version 0.4 applies Amish's decisions of 2026-09-25 (WWT-DDR-002): R1 is relaxed, R11 is restated for a footing cast on the survey visit, R5 gains a 15 min confirming reading and R12 now states that the pH probe variant is priced per variant site. The targets are still not user-validated needs and will be revised after co-design. On paper, no requirement is missed outright; four are at risk (R1, R10, R2 and R9), five are met by calculation and three by design, and the parts cost of the constructable design (WWT-DDR-003) is USD 19 over the USD 300 value-engineering target (R12). Version 0.5 updates the status for that design; no target changed. In v0.3, R1 and R10 were not met and R11 was at risk. Amish's decisions are in WWT-DDR-001 and WWT-DDR-002.

Table 1. Requirements and TRL 3 status (from WWT-CAL-001 v0.4, Table 4), at-risk items first.

ID	Requirement	Target	Verification	TRL 3 status
R1	Measure free chlorine at the tap	0 to 2.0 mg/L; within ± 0.2 mg/L or ± 25 %, whichever is greater, at pH 6.5 to 8.5 and 5 to 40 °C (relaxed from ± 0.1 mg/L below 1.0 mg/L and ± 15 % above, WWT-DDR-002). Sites whose pH is above 7.5 or moves more than 0.2 between visits use the pH probe variant	Error budget (WWT-CAL-001, E); later side-by-side with a DPD comparator	At risk: met in 8 of 8 cases at 25 °C before drift, with at least 0.13 mg/L left for drift; 1.5 mg/L at pH 7.5 and 40 °C misses by 0.006 mg/L; sensor drift unknown
R10	Be maintainable by a caretaker	One visit per month or less, 30 min or less, no special tools: clean the cell, compare with a DPD kit, adjust calibration from a phone or keypad	Task analysis (WWT-CAL-001, J)	At risk: the visit takes 30 min, at the limit, and a monthly interval needs chlorine drift below about 0.13 mg/L per month under the relaxed R1 (was 0.04 mg/L), which no verified data show
R2	Measure turbidity	0 to 100 NTU, reporting "above 100 NTU" beyond; within ± 0.5 NTU or ± 10 %, whichever is greater	Published open-source design; bubble and settling calculation (WWT-CAL-001, F); later stabilized standards	At risk: the 30 s wait clears 100 μ m bubbles at 5 °C; window fouling between cleans is unknown
R9	Survive outdoors	Electronics IP65 or better; ambient 0 to 45 °C in direct sun; UV-stable parts; flow cell opaque to daylight; battery charging blocked below 0 °C and above 45 °C	Thermal estimate (WWT-CAL-001, C); parts selection	At risk: with the sun shield the inside peaks at 50.8 °C on a 45 °C day (69.8 °C without); the shield factor is assumed
R5	Alert the people who act	SMS to up to three numbers within 15 min of a confirmed crossing (two consecutive readings, the second taken 15 min after the first crossing, WWT-DDR-002): free chlorine below 0.2 mg/L, turbidity above 5 NTU, device fault or low battery; thresholds adjustable per site	Latency calculation (WWT-CAL-001, H)	Met on paper where there is coverage: 13.0 min worst case with three tries at 3 min spacing; 77 min from the event (was 122 min)
R6	Report and keep data	Upload readings at least every 4 h over LTE-M, NB-IoT or 2G (LoRaWAN variant); keep 90 days on board if the link fails; open CSV or JSON format; works without a cloud service	Data budget (WWT-CAL-001, G)	Met on paper: 0.86 MB per month; 90 days in 138 kB
R7	Run on sunlight alone	7 days or more with no sun from full; back to full in 3 clear days or fewer	Energy budget (WWT-CAL-001, A and B)	Met on paper: 28.5 days (24.2 at 0 °C); 1.19 clear days to refill; the status light costs 2 mWh a day; hot days rely on the sun shield (R9)
R8	Use little treated water	20 L/day or less at hourly sampling	Flow calculation (WWT-CAL-001, D)	Met on paper, thin margin: 18.0 L/day nominal, 19.8 L at the regulator's tolerance

ID	Requirement	Target	Verification	TRL 3 status
R11	Install simply and safely	Two people, 2 h or less on installation day, with the pole footing dug and cast on the survey visit (WWT-DDR-002); hand tools only; one tee and isolation valve on the riser; backflow prevented; wetted parts food-safe	Task analysis (WWT-CAL-001, J); design review	Met on paper: 65 min critical path on installation day (140 min if done in one visit); the design parts are met
R12	Stay within the concept budget	Parts cost per base unit against a value-engineering target of \$300 (a hypothetical control target, not a limit), excluding the shared calibration kit, airtime and the pH probe variant, which is priced per variant site (WWT-DDR-002)	Priced BOM (WWT-CAL-001, K)	Over the value-engineering target by \$22: \$322 per base unit for the constructable design; \$382 to \$422 at a pH probe variant site
R3	Measure water temperature	0 to 50 °C, within ± 0.5 °C	Sensor datasheet	Met by design
R4	Sample on a schedule	One reading per hour by default; configurable from 15 min to 24 h	Energy at the 15 min schedule (WWT-CAL-001, A and B)	Met by design (15 min schedule: 1.589 Wh/day, 9.7 days on battery, but 72 L/day of water)
R13	Report measurements, not verdicts	Messages state what was measured and when; the device never says the water is "safe"	Design review of alert wording	Met by design

■ Assumptions

- Site: public tapstand on a piped, chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi) at the riser. Tapstands come first; tank outlets and kiosks later (WWT-DDR-001, D8).
- Alert thresholds follow the field guidance summarized in WWT-PRB-001: at least 0.2 mg/L free chlorine at the point of delivery, and chlorination losing effectiveness as turbidity rises. The defaults of 0.2 mg/L and 5 NTU were decided by Amish on 2026-09-25 (WWT-DDR-001, D6) and remain adjustable per site.
- The caretaker's DPD test is the field reference for free chlorine. The v0.3 target of ± 0.1 mg/L matched the resolution of common DPD comparator kits, which WWT-CAL-001 showed made it as tight as the reference itself (the comparator alone takes ± 0.07 mg/L); R1 was relaxed to ± 0.2 mg/L or ± 25 % for that reason (WWT-DDR-002).
- pH affects the split between hypochlorous acid and hypochlorite, and amperometric sensors respond mainly to hypochlorous acid. R1 is assessed with a site pH entered at installation and checked monthly (WWT-DDR-001, D2), assumed to move no more than ± 0.2 between visits. Under WWT-DDR-002 the pH probe variant (read to ± 0.1) is fitted at sites whose pH is above 7.5 or moves more than 0.2 between visits; the cell lid carries a plugged spare port for it.
- Budget: the \$300 in `project.yaml` is a value-engineering target for one base unit's parts, not a spending limit (Amish, 2026-10-01). A shared calibration kit (about \$60), airtime (about \$1 to \$3 per month) and, at variant sites only, the pH probe and interface (about \$60 to \$100) are listed separately; all are estimates.
- Installation: the pole footing is dug and cast on the survey visit, so the pole is set and cured before installation day (WWT-DDR-002).

Safety: Meeting these requirements does not make the water safe to drink or the unit safe to install. The sample line connects to a pressurized drinking water supply and the unit contains a lithium iron phosphate battery; see WWT-PRC-001, Safety.